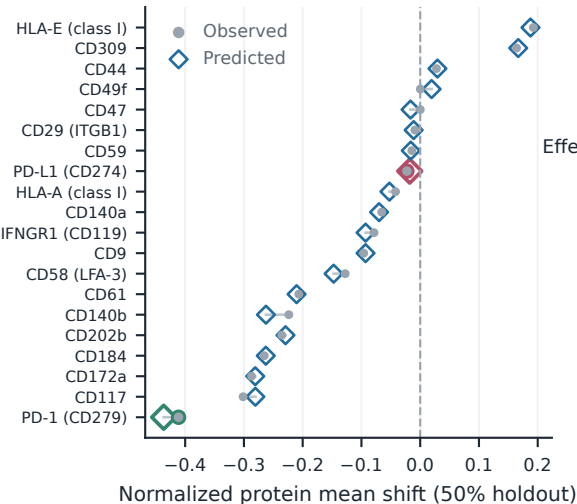
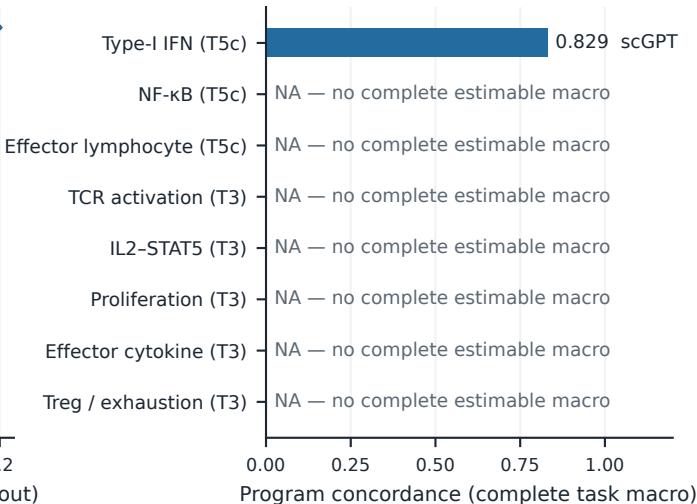


# Immune readouts retain distinct marker, program and lineage information

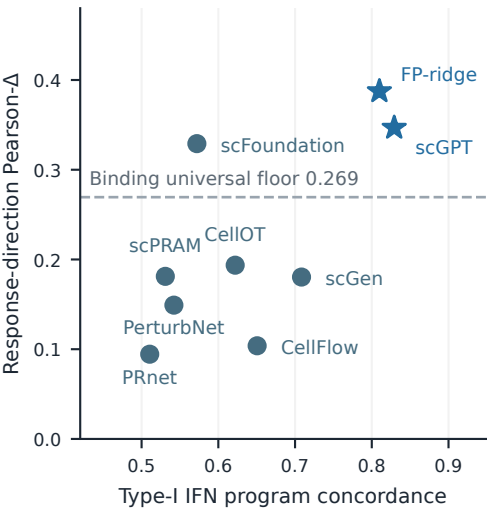
**a Surface-marker shifts**



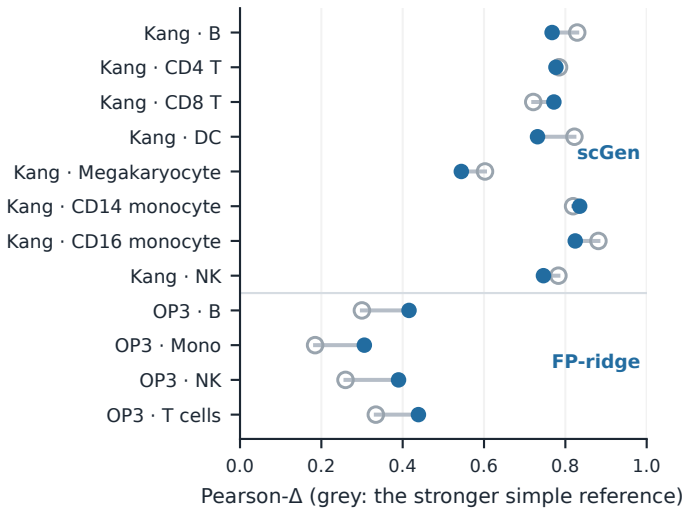
**b Immune-program concordance**



**c Program and transcriptome agreement**



**d Prediction varies by lineage**



Protein reference is fitted within modality. Undefined program correlations are not zero recovery. Local point advantages do not establish task-wide or multiplicity-adjusted support.